CPEN 416

Lecture 14: QPE and order finding

Thursday 29 October 2026

Announcements

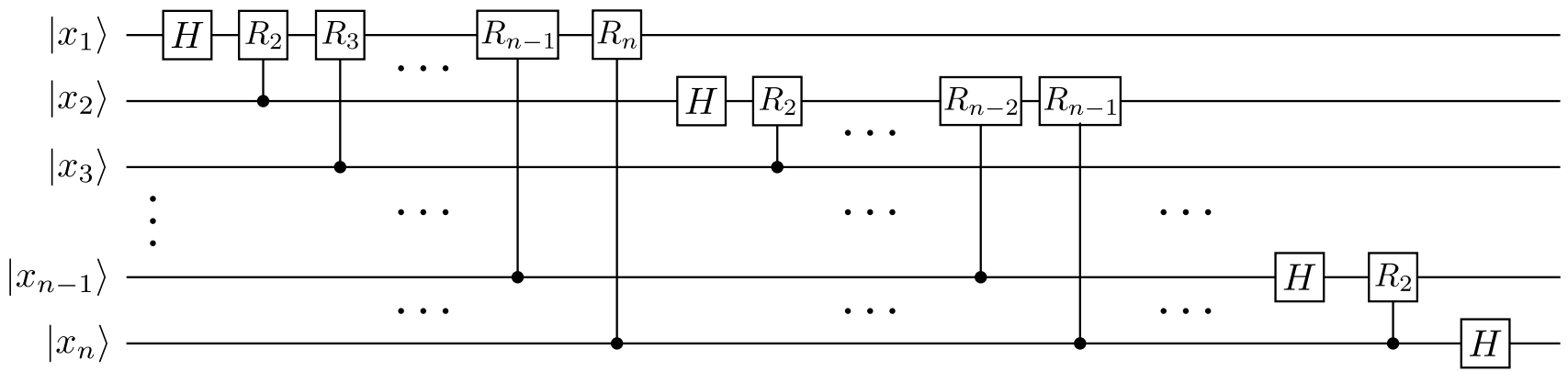
- Quiz 7 on Tuesday
- Tutorial Assignment 2 due Friday 23:59
- Midterm checkpoint report due Monday **9:00am**
  - Calendar bookings for checkpoints meetings available
- Practice problem set 3 available (about QPE)

- Monday tutorial: mini lecture about RSA

Last time

We implemented the quantum Fourier transform using a circuit with a *polynomial* number of gates and depth.

$$QFT|x\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{xk} |k\rangle$$



Reminder: where are we going?

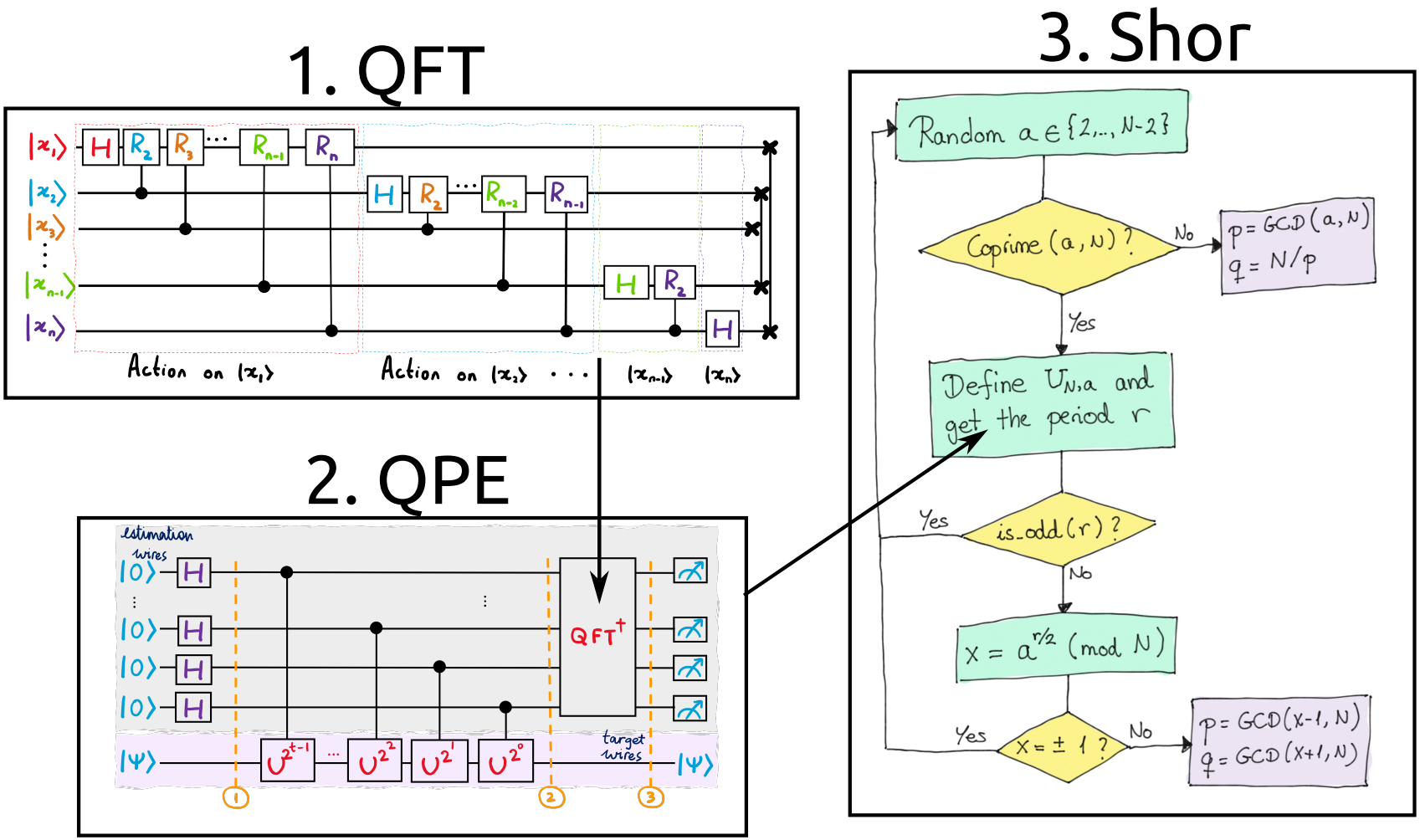


Image credit: Xanadu Quantum Codebook nodes F.3, P.3, S.4

Learning outcomes

- Implement the quantum phase estimation (QPE) subroutine
- Perform arithmetic modulo N with integers and with quantum operations
- Define the order finding problem
- Apply QPE to solve the order finding problem

Eigenvalues of unitary matrices

$U = e^{-i\hat{H}t}$ → $\hat{H}$: Hermitian matrix; in physics: Hamiltonian

Eigenvalues of unitary matrices are complex numbers with magnitude 1, i.e.,

$|k\rangle$ is eigenvector (eigenstate) of U: $U|k\rangle = \lambda_k |k\rangle$

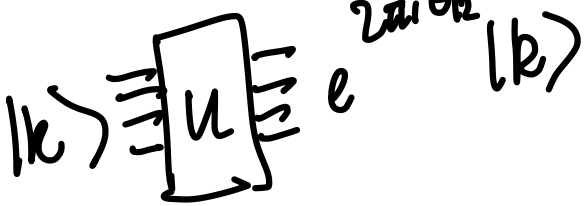
$\lambda_k = e^{2\pi i \theta_k} = e^{i\phi_k}$

where θ_k is a phase, $|\theta_k| \leq 1$. (See proof at end of slides)

$\theta_k = 0.\theta_{k_1}\theta_{k_2}\cdots$

What if we want to *estimate* an unknown θ_k?

$U|k\rangle = e^{2\pi i \theta_k}|k\rangle$



Quantum phase estimation

Given unitary U and eigenvector $|k\rangle$, estimate θ_k such that

$$U|k\rangle = e^{2\pi i\theta_k}|k\rangle$$

apply lots of phase kickback

Must determine:

- How to design a circuit that extracts θ_k
- What precision can we estimate it to
- What to do if we don't know $|k\rangle$ in advance

(*You can explore the last two in practice assignment 3!*)

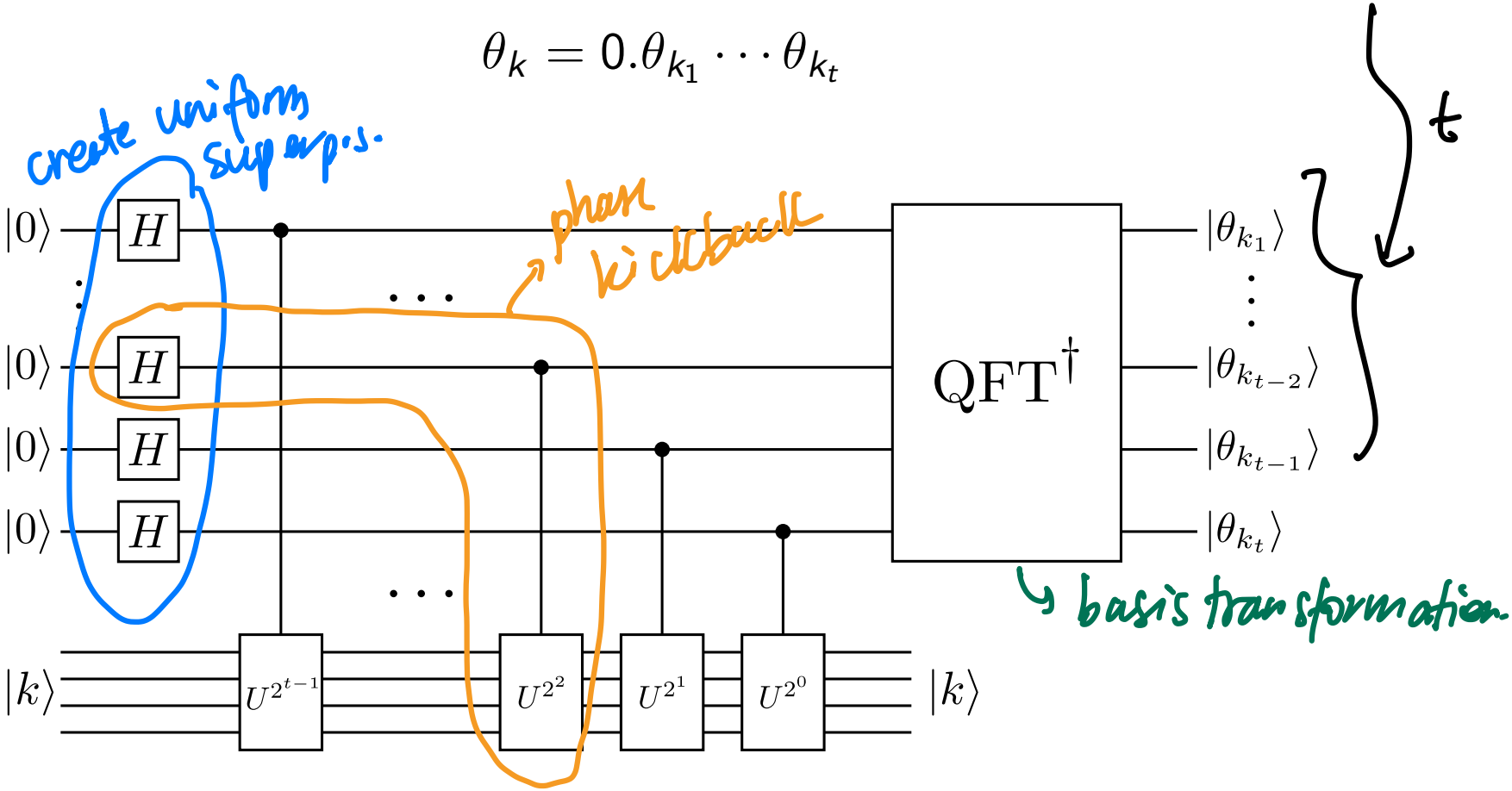
Quantum phase estimation

Let U be an n-qubit unitary; $|k\rangle$ is an n-qubit eigenstate.

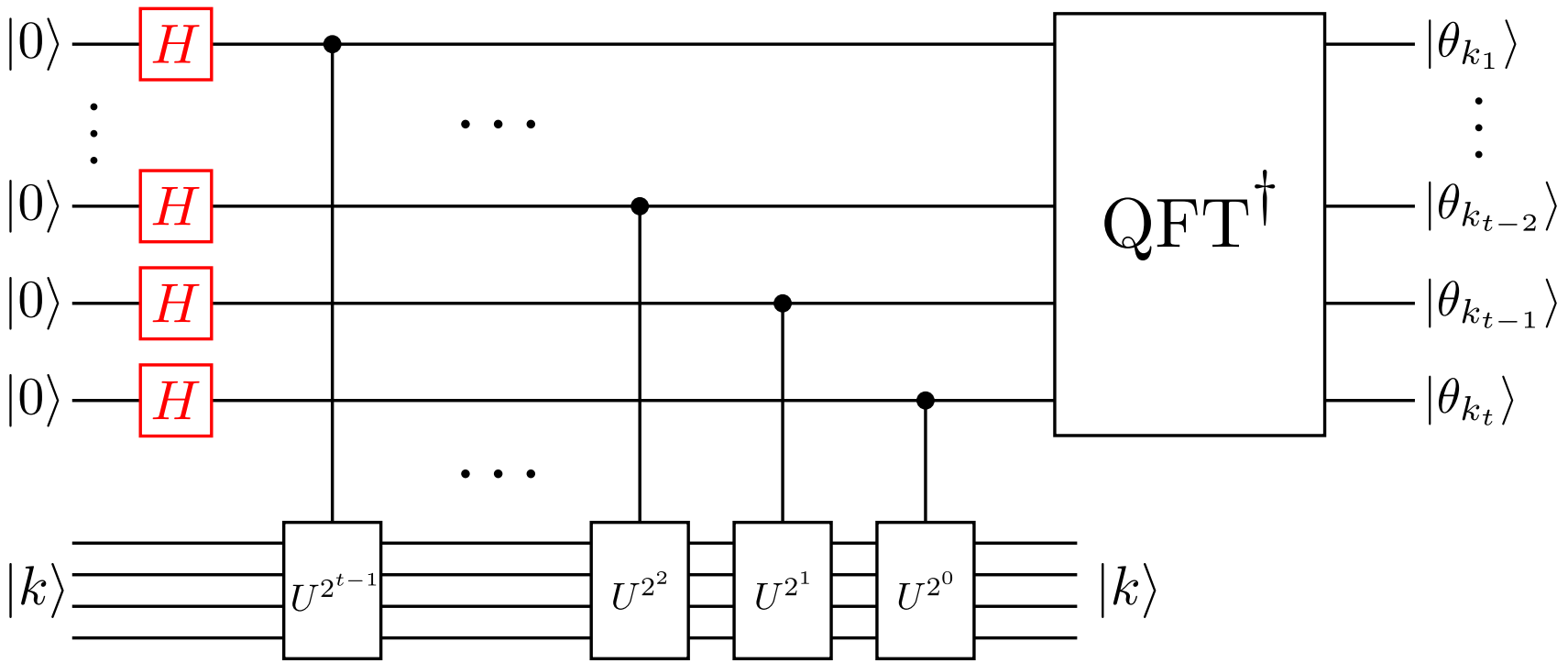
$\theta_k = 0.8125 = \underbrace{0.1101}_{4 \text{ bits}}$

Assume θ_k can be represented *exactly* using t bits:

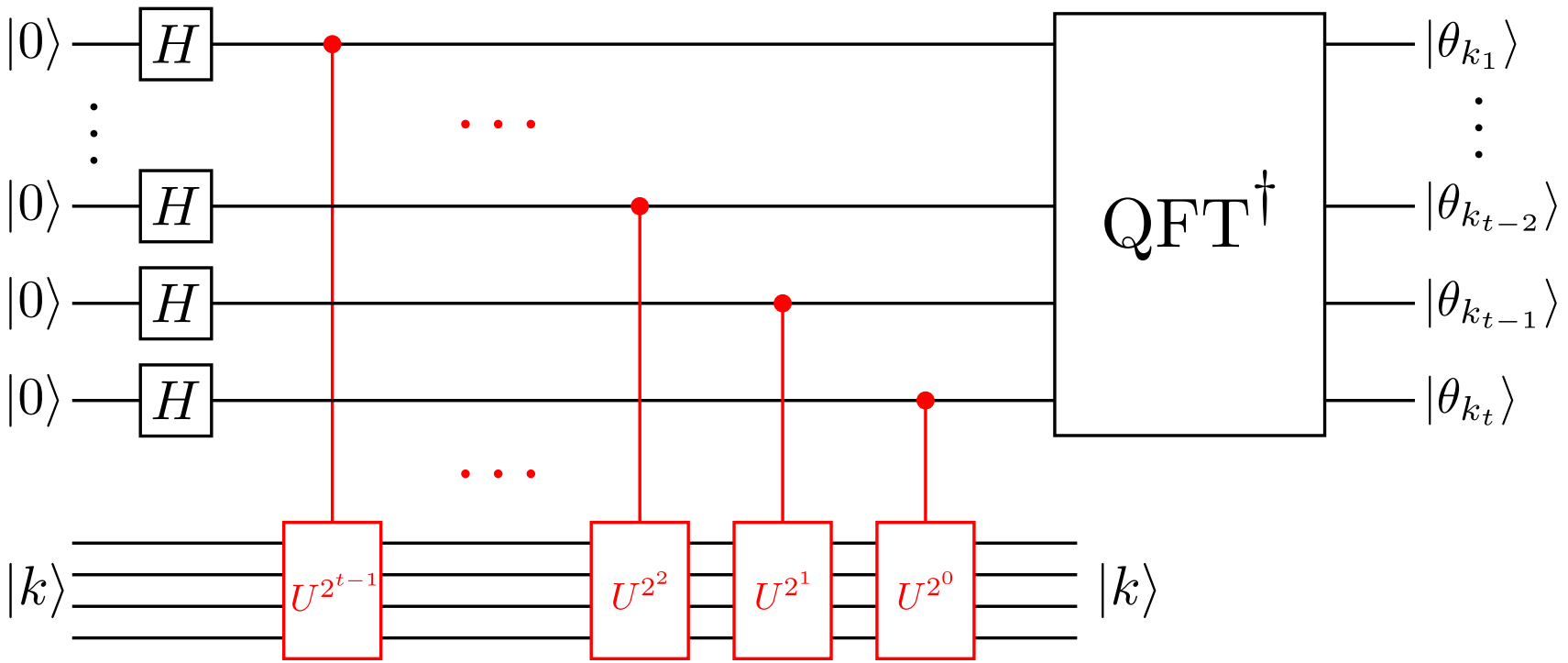
$$\theta_k = 0.\theta_{k_1} \cdots \theta_{k_t}$$



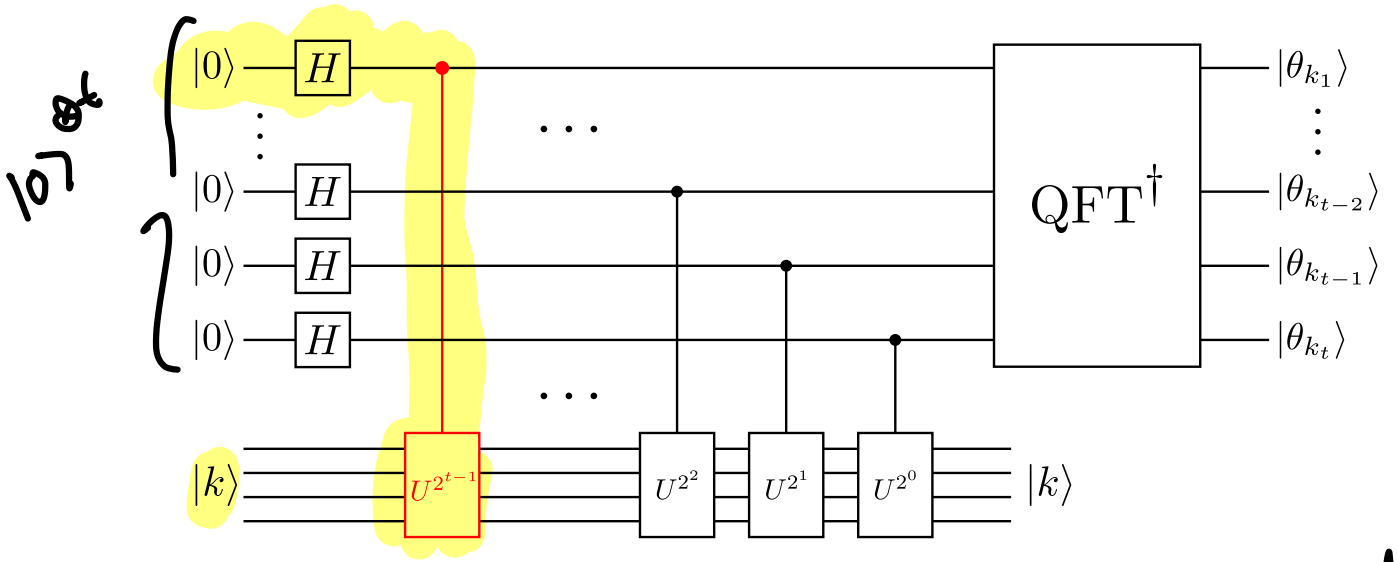
Quantum phase estimation: step 1



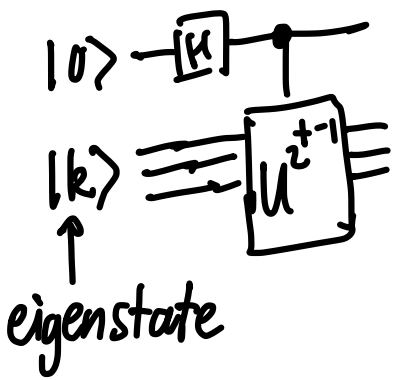
Quantum phase estimation: step 2



Quantum phase estimation: step 2



$|0\rangle^{\otimes t}$



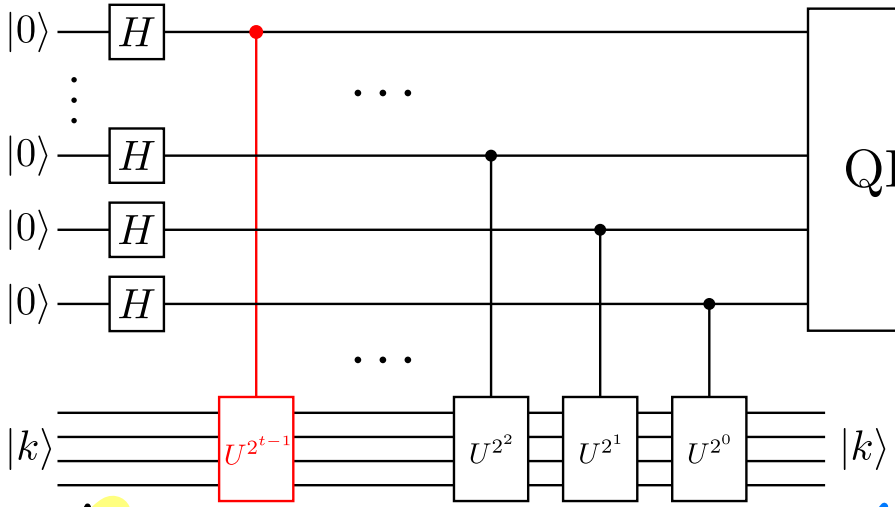
$$|0\rangle|k\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)|k\rangle = \frac{1}{\sqrt{2}}|0\rangle|k\rangle + \frac{1}{\sqrt{2}}|1\rangle|k\rangle$$

Apply controlled $U^{2^{t-1}}$:

$$\frac{1}{\sqrt{2}}|0\rangle|k\rangle + \frac{1}{\sqrt{2}}|1\rangle\left(U^{2^{t-1}}|k\rangle\right) = \frac{1}{\sqrt{2}}|0\rangle|k\rangle + \frac{1}{\sqrt{2}}|1\rangle\left(e^{2\pi i\theta_k}\right)^{2^{t-1}}|k\rangle$$

$$= \frac{1}{\sqrt{2}}\left(|0\rangle + \left(e^{2\pi i\theta_k}\right)^{2^{t-1}}|1\rangle\right)|k\rangle$$

Quantum phase estimation: step 2



$e^{2\pi i} = 1$

$e^{2\pi i \cdot 4} = 1$

$\theta_k = 0.\theta_{k_1}\theta_{k_2}\cdots\theta_{k_t}$

$\frac{1}{\sqrt{2}}\left(|0\rangle + \left(e^{2\pi i \theta_k}\right)^{2^{t-1}}|1\rangle\right)|k\rangle$

$\left(e^{2\pi i \theta_k}\right)^{2^{t-1}} = e^{2\pi i \theta_k \cdot 2^{t-1}}$

$2\pi i \left[\frac{\theta_{k_1}}{2} + \frac{\theta_{k_2}}{2^2} + \cdots \frac{\theta_{k_t}}{2^t}\right] \cdot 2^{t-1}$ ($\theta_{k_1}, \theta_{k_2}, \ldots$: 0 or 1)

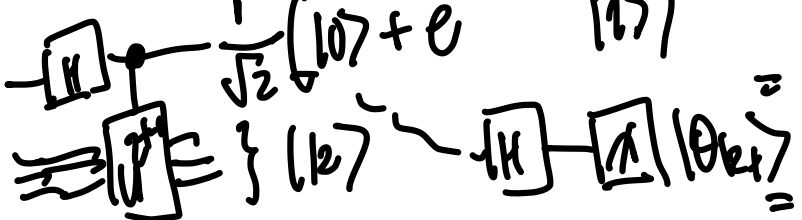
$= e^{2\pi i \left[\frac{\theta_{k_1}\cdot 2^{t-1}}{2} + \frac{\theta_{k_2}\cdot 2^{t-1}}{2^2} + \cdots + \frac{\theta_{k_t}\cdot 2^{t-1}}{2^t}\right]}$

$= e^{2\pi i \left[\theta_{k_1}\cdot 2^{t-2} + \theta_{k_2}\cdot 2^{t-3} + \cdots \theta_{k_t}\cdot 2^{-1}\right]}$

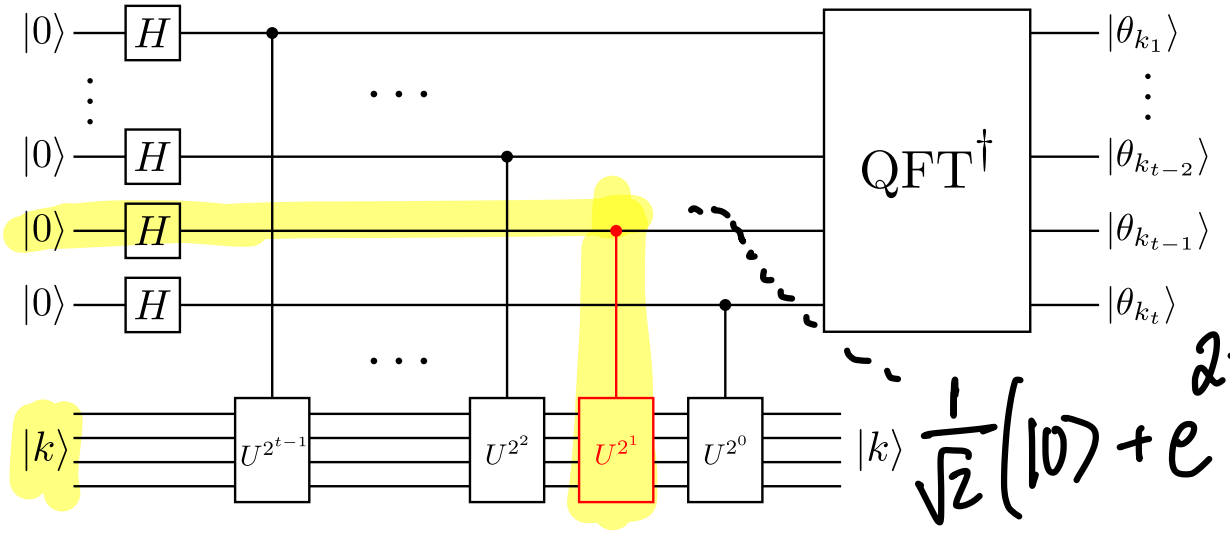
$\theta_{k_1}\cdot 2^{t-2}$: integer, $e^{2\pi i} = 1$; $\theta_{k_2}\cdot 2^{t-3}$: integer, $= 1$; $\theta_{k_t}\cdot 2^{-1}$: not just

$= e^{2\pi i \cdot \theta_{k_t}/2} = e^{2\pi i \cdot 0.\theta_{k_t}}$

$\frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i 0.k_t}|1\rangle\right)$



Quantum phase estimation: step 2



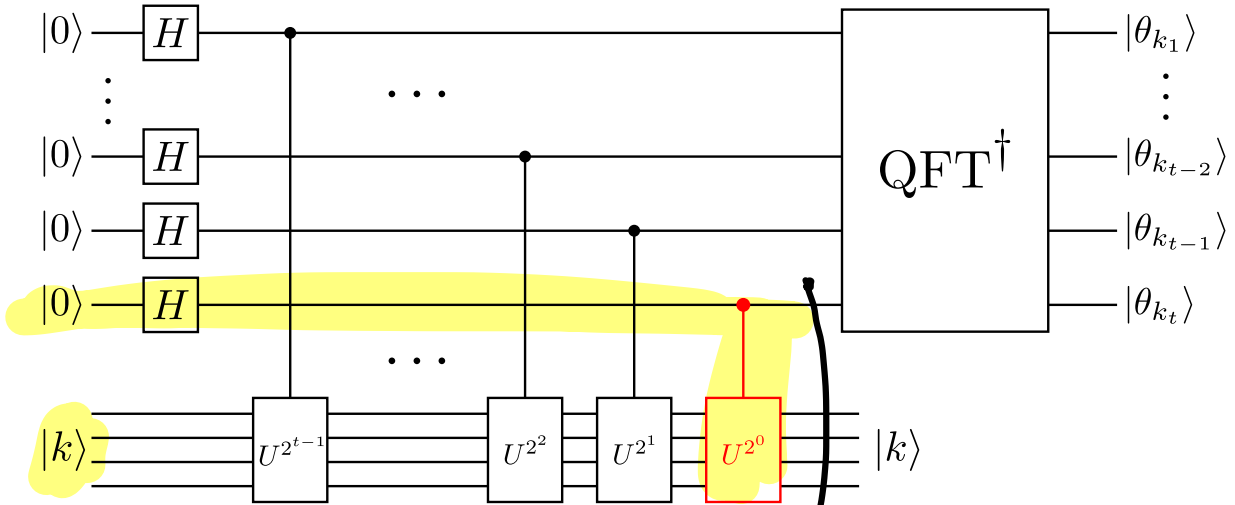
Check second-last qubit (ignore the others)

$$\frac{1}{\sqrt{2}}|0\rangle|k\rangle + \frac{1}{\sqrt{2}}|1\rangle|k\rangle \xrightarrow{c\text{-}U^2} \frac{1}{\sqrt{2}}\left(|0\rangle + \left(e^{2\pi i \theta_k 2}\right)|1\rangle\right)|k\rangle$$

$$\left(e^{2\pi i \theta_k}\right)^2 = e^{2\pi i \theta_k \cdot 2} = e^{2\pi i\left(\frac{\theta_{k_1}}{2} + \frac{\theta_{k_2}}{2^2} + \cdots \frac{\theta_{k_t}}{2^t}\right)\cdot 2}$$

$$= e^{2\pi i\left(\theta_{k_1} + \frac{\theta_{k_2}}{2} + \frac{\theta_{k_3}}{2^2} + \cdots \frac{\theta_{k_t}}{2^{t-1}}\right)} = e^{2\pi i\left(\frac{\theta_{k_2}}{2} + \frac{\theta_{k_3}}{2^2}\cdots\right)}$$

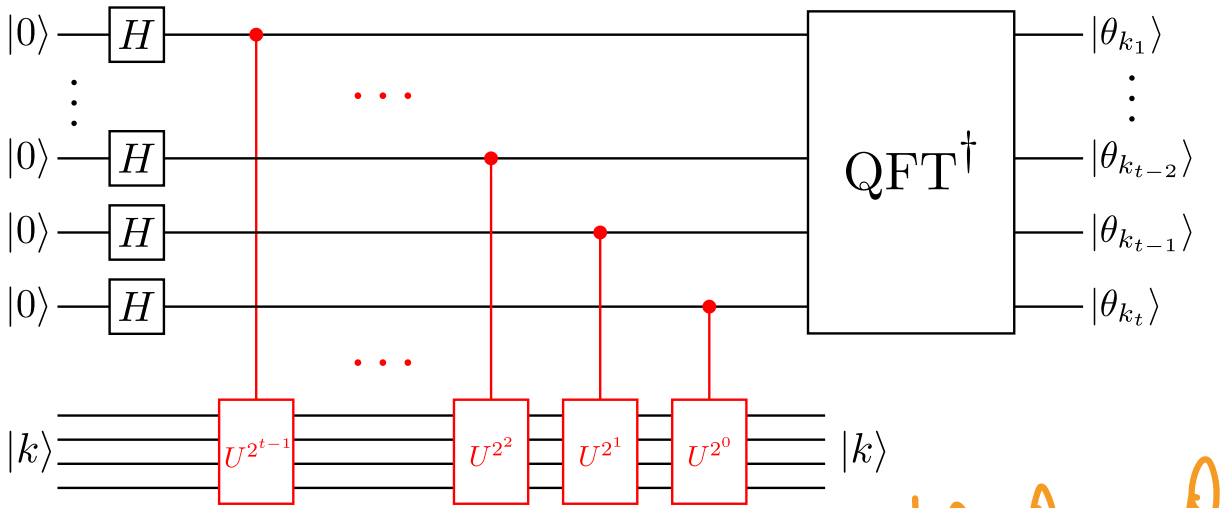
Quantum phase estimation: step 2



Can show in the same way for the last qubit (ignore others)

$$\frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i \cdot 0.\theta_t\theta_{t-1}\cdots\theta_1}|1\rangle\right)$$

Quantum phase estimation: step 2



$\mathrm{QFT}\,|\theta_{k_1}\theta_{k_2}\cdots\theta_{k_t}\rangle$

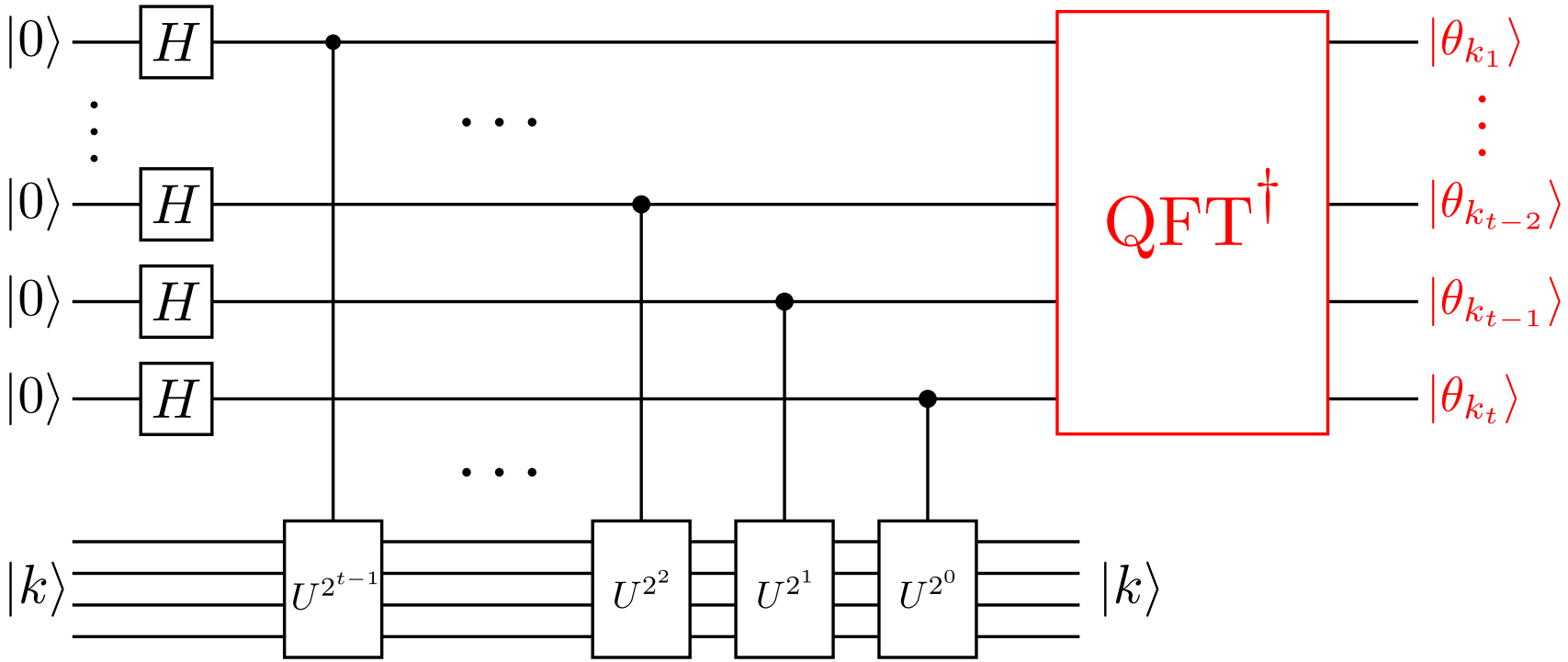
After step 2, we have the state

$$\frac{1}{\sqrt{2}}(|0\rangle+e^{2\pi i 0.\theta_{k_t}}|1\rangle)\cdots\frac{1}{\sqrt{2}}(|0\rangle+e^{2\pi i 0.\theta_{k_2}\cdots\theta_{k_t}}|1\rangle)\frac{1}{\sqrt{2}}(|0\rangle+e^{2\pi i 0.\theta_{k_1}\cdots\theta_{k_t}}|1\rangle)|k\rangle$$

Should look familiar!

Quantum phase estimation: step 3

Measure to learn the bits of θ_k.



Example: QPE for the T gate

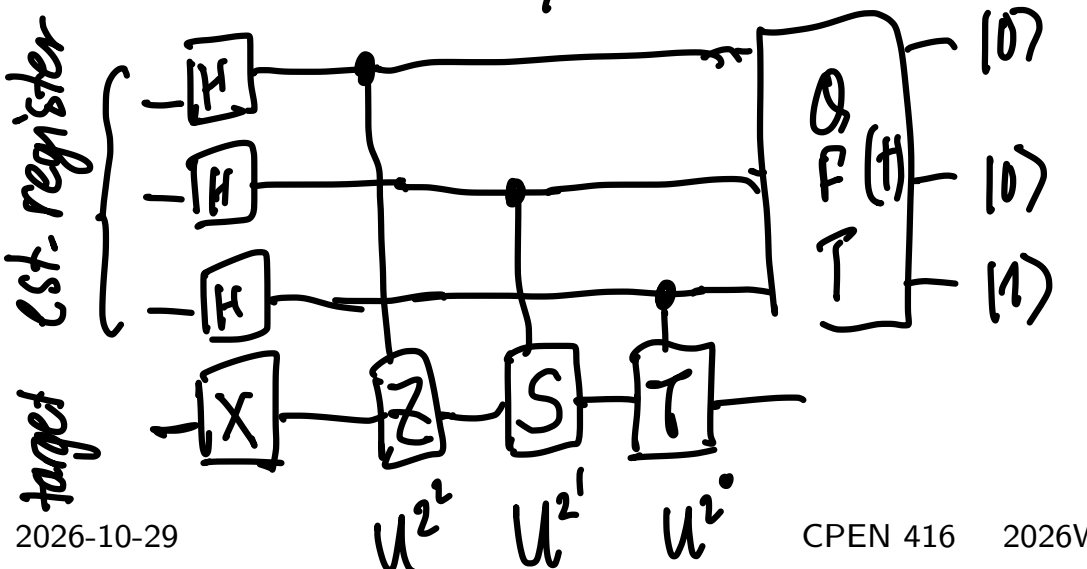
Let's apply QPE to estimate the phase of an eigenstate of T: $|1\rangle$.

1. What answer do we expect?
2. How many estimation bits?
3. What does the circuit look like?

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} \qquad T|1\rangle = e^{i\pi/4}|1\rangle$$

$$\lambda_k = e^{i\pi/4} = e^{2\pi i \theta_k}$$

$$\theta_k = \frac{1}{8} \Rightarrow 0.001 \quad t = 3$$



Quantum phase estimation

Recall our initial questions... any insights about the second two?

- How to design a circuit that extracts θ_k
- What precision can we estimate it to
- What to do if we don't know $|k\rangle$ in advance

Modular arithmetic

Consider the finite set of integers $\{0, 1, \ldots, N-1\}$.

Basic arithmetic modulo N:

$5 \cdot 8 \mod 13 = 1$

Modular inverses

The inverse of a modulo N (a^{-1}) is an integer b such that

$$a \cdot b = a a^{-1} \equiv 1 \mod N$$

equiv. to 1 modulo N

A modular inverse only exists when a and N are *coprime*.

4, 8 ⇒ not coprime 3, 8 ⇒ coprime.

Example: $N = 5$

x	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

Example: $N = 4$

x	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

Modular exponentiation

Often simplified using techniques such as repeated squaring.

Example: evaluate $8^6 \bmod 11$

Method 1:

Method 2:

8^2

8^3

8^4

$\vdots$

Method 3:

Modular arithmetic in PennyLane

PennyLane contains *built-in* operations:

- `qp.Adder`: $|x\rangle \to |x + y \bmod N\rangle$
- `qp.OutAdder`: $|x\rangle|y\rangle|b\rangle \to |x\rangle|y\rangle|b + x + y \bmod N\rangle$
- `qp.Multiplier`: $|x\rangle \to |xy \bmod N\rangle$
- `qp.OutMultiplier`: $|x\rangle|y\rangle|b\rangle \to |x\rangle|y\rangle|b + xy \bmod N\rangle$
- `qp.ModExp`: $|x\rangle|y\rangle \to |x\rangle|y \cdot a^x \bmod N\rangle$

These implementations actually leverage the QFT under the hood; see explanation in this PennyLane demo*.

*https://pennylane.ai/qml/demos/tutorial_shors_algorithm_catalyst#appendix-fun-with-fourier-transforms

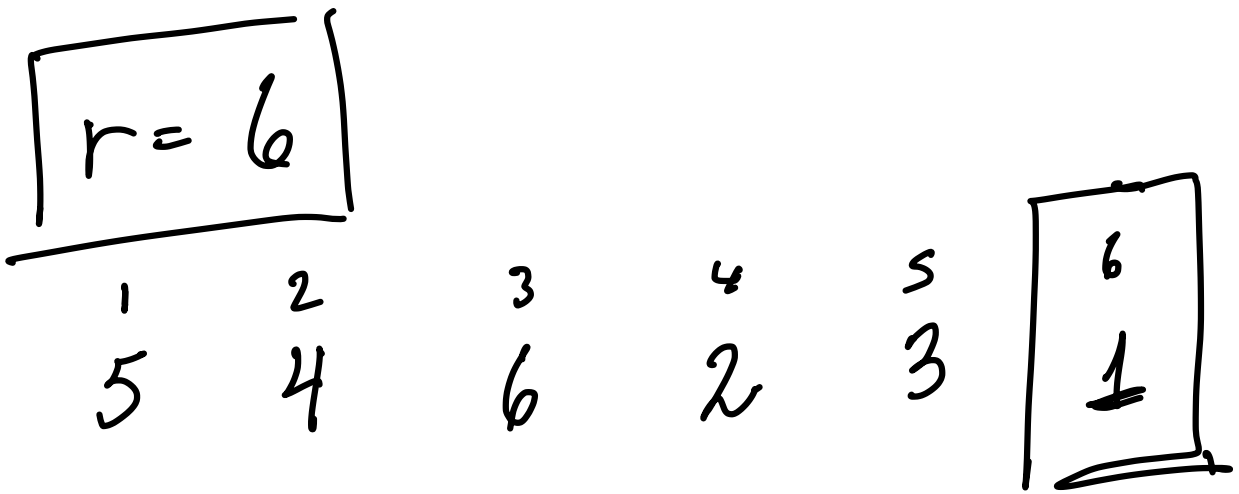
Order finding

Input: integers a and N, where $N \geq 2$ and $\gcd(a, N) = 1$
Output: smallest integer r for which $a^r \equiv 1 \bmod N$

Exercise: find the order of $a = 5$ for $N = 7$.

$5^r \equiv 1 \bmod 7$

??



Order finding vs. period finding

Consider the function

Order finding is equivalent to *period finding* for $f_{N,a}(x)$:

Order finding on a quantum computer

$$f_{N,a}(x) = a^x \bmod N$$

Define a unitary operation that performs

$$U_{N,a}|k\rangle = |ak \bmod N\rangle$$

If r is the order of a modulo N, and we apply $U_{N,a}$ r times,

$$\left(U_{N,a}\right)^r |k\rangle = |a^r k \bmod N\rangle = |k \bmod N\rangle$$

Order finding on a quantum computer

$U_{N,a}$ has the same order, r. We can use QPE to efficiently find r with a clever initialization of the target register.

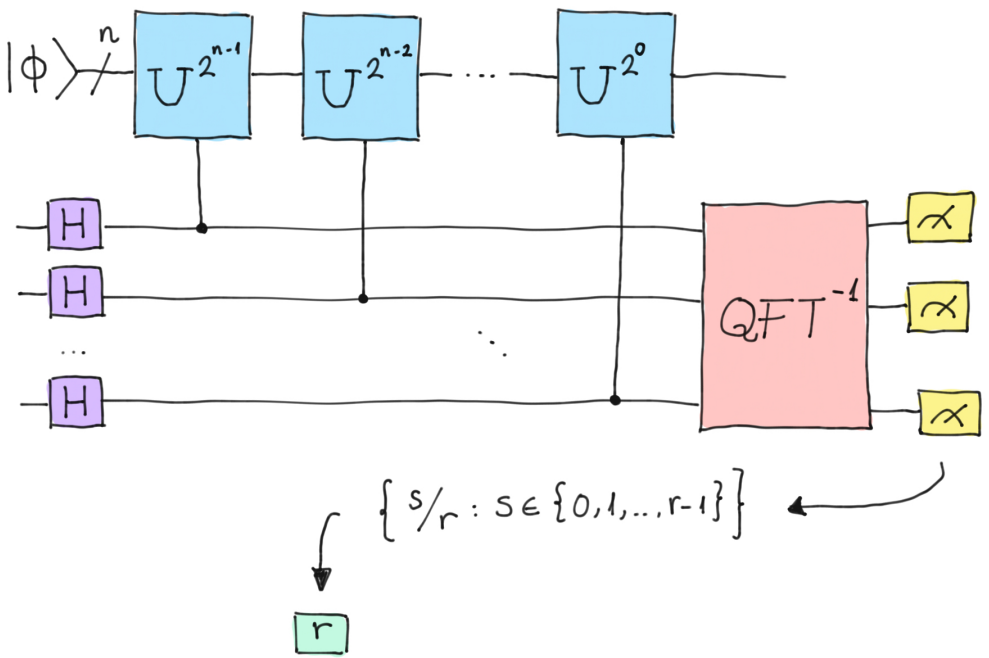


Image credit: Xanadu Quantum Codebook node S.3

Next time

Content:

- Factoring with Shor's algorithm

Action items:

1. Tutorial Assignment 2 due tomorrow 23:59
2. Midterm checkpoint report due Monday **9:00am**

Recommended reading:

- Codebook modules QFT, QPE, SH
- Nielsen & Chuang 5.3, Appendix A.5

Eigenvalues of unitary matrices

Fun fact: eigenvalues of unitaries are complex numbers with magnitude 1.

Proof: Let λ_k be the eigenvalue associated with eigenvector $|k\rangle$ of a unitary U:

$$U|k\rangle = \lambda_k|k\rangle$$

Take the conjugate transpose:

$$\langle k| U^\dagger = \langle k| \lambda_k^*$$

Multiply the two sides together:

$$\langle k| U^\dagger U|k\rangle = \langle k| \lambda_k^* \lambda_k|k\rangle$$
$$\langle k|k\rangle = |\lambda_k|^2 \langle k|k\rangle$$
$$1 = |\lambda_k|^2$$